

SANTA MARIA AP, PortugalWMO: **085150**Lat: **36.973N**Lon: **25.173W**Elev: **318**StdP: **14.53**Time Zone: **-1.00 (AZR)**Period: **94-19**WBAN: **99999****Annual Heating, Humidification, and Ventilation Design Conditions**

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	51.3	52.1	40.7	37.7	54.7	42.4	40.4	55.5	35.8	59.3	32.5	58.7	11.6	60	0.459

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	9.1	79.1	71.5	78.4	71.1	77.0	70.2	73.7	76.9	72.8	76.1	72.0	75.4	11.1	300

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
72.5	122.0	75.5	71.7	118.4	75.0	70.5	113.8	74.2	37.4	77.4	36.5	75.9	35.8	75.3	81.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
30.9	27.2	24.1	DB	47.8	81.3	1.6	1.2	46.7	82.2	45.7	82.9	44.8	83.6	43.7	84.5	
			WB	44.2	75.6	1.5	1.6	43.1	76.8	42.3	77.7	41.4	78.6	40.4	79.8	

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
Temperatures, Degree-Days and Degree-Hours	DBAvg	64.8	59.4	58.6	59.2	60.6	63.2	67.1	71.0	73.2	71.8	68.2	63.6	61.0	
	DBStd	5.69	2.72	2.50	2.46	2.32	2.50	2.45	2.40	2.07	2.31	2.92	3.13	2.85	
	HDD50	0	0	0	0	0	0	0	0	0	0	0	0	0	
	HDD65	939	174	179	180	134	68	9	0	0	0	7	63	125	
	CDD50	5394	291	241	285	317	409	512	652	720	653	564	408	342	
	CDD65	856	0	0	0	0	12	70	187	255	203	105	21	3	
	CDH74	1404	0	0	0	0	0	18	277	700	363	45	1	0	
	CDH80	17	0	0	0	0	0	0	4	12	1	0	0	0	
Wind	WSAvg	11.7	13.4	13.1	13.7	12.7	11.2	9.9	8.7	8.9	9.7	12.6	12.5	14.2	
Precipitation	PrecAvg	29.1	3.5	2.9	3.0	2.1	1.3	1.0	0.8	1.3	2.1	3.7	3.5	3.9	
	PrecMax	51.3	11.3	8.8	10.3	5.5	5.2	3.5	4.4	6.3	6.3	10.2	13.2	11.5	
	PrecMin	17.5	0.1	0.1	0.0	0.2	0.0	0.0	0.0	0.2	0.2	0.7	0.3	0.6	
	PrecStd	6.5	2.4	2.0	2.2	1.6	1.1	0.9	0.8	1.2	1.3	2.4	2.5	2.3	
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	66.4	66.2	66.6	68.6	71.9	75.6	80.2	81.0	79.8	77.1	73.3	68.9	
		MCWB	62.6	61.5	60.2	61.9	65.3	68.0	71.6	72.3	72.0	70.7	68.0	64.4	
	2%	DB	64.8	64.6	64.8	66.6	70.1	73.8	78.3	79.9	78.6	75.3	71.2	67.4	
		MCWB	61.6	60.5	59.7	61.1	64.2	67.0	70.9	71.8	71.5	69.2	66.9	63.8	
	5%	DB	64.3	63.3	64.3	65.9	68.7	73.0	76.9	78.8	77.2	73.7	69.7	66.3	
		MCWB	61.2	59.7	59.4	60.6	63.2	66.5	69.9	71.3	70.4	68.4	65.7	63.2	
	10%	DB	63.0	62.6	62.9	64.5	67.8	71.6	75.4	77.3	76.2	72.9	68.2	65.1	
		MCWB	60.2	59.2	58.7	59.6	62.5	65.6	68.8	70.4	69.7	68.0	64.5	62.2	
	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	63.9	63.6	63.4	64.2	67.7	71.0	73.8	75.4	74.7	72.9	69.6	66.2
			MCDB	65.1	64.8	64.8	66.5	70.0	73.8	77.1	78.4	77.5	75.1	71.7	67.6
2%		WB	63.2	62.1	62.0	63.2	66.2	69.3	72.6	74.0	73.5	71.2	68.2	65.1	
		MCDB	64.5	63.5	63.6	65.4	68.5	71.9	76.2	77.4	76.4	73.5	70.0	66.6	
5%		WB	62.1	61.3	61.3	62.1	65.1	68.4	71.6	72.9	72.6	70.2	67.0	63.9	
		MCDB	63.6	62.8	63.0	64.4	67.6	71.2	75.1	76.5	75.5	72.6	68.9	65.5	
10%		WB	61.5	60.1	60.2	61.4	63.8	67.2	70.6	72.2	71.5	69.1	65.6	63.3	
		MCDB	63.1	61.9	62.2	63.7	66.5	70.2	74.1	75.9	74.6	71.6	67.7	64.9	
Mean Daily Temperature Range		5% DB	MDBR	6.4	7.2	7.6	8.0	8.3	8.6	9.1	9.1	9.0	7.7	7.2	6.4
			MCDBR	6.3	7.5	8.2	8.7	8.9	9.3	9.7	9.7	9.5	8.4	7.7	6.8
	MCWBR		5.4	6.0	6.2	5.9	5.9	5.6	5.4	5.5	5.7	5.9	5.8	6.0	
	5% WB	MCDBR	5.5	6.6	6.9	7.6	7.6	7.9	8.5	8.5	8.2	7.4	6.5	6.1	
		MCWBR	5.8	6.4	6.3	6.0	6.2	6.2	5.9	6.1	6.5	6.4	6.2	6.3	
Clear-Sky Solar Irradiance	taub	0.345	0.351	0.372	0.374	0.368	0.369	0.363	0.361	0.359	0.368	0.350	0.351		
	taud	2.547	2.530	2.466	2.470	2.514	2.548	2.571	2.599	2.610	2.563	2.557	2.524		
	Ebn at Noon	268	281	283	288	290	288	289	288	285	271	264	257		
	Edh at Noon	25	28	33	35	34	32	31	30	28	27	24	24		
All-Sky Solar Radiation	RadAvg	717	980	1341	1710	1975	2094	2092	1977	1617	1132	795	612		
	RadStd	57	60	89	112	120	123	110	81	69	65	52	49		

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
Station Only	N/A	N/A	N/A	N/A	+0.54	+0.79	N/A	N/A	N/A	N/A
Regional (1 neighbor)	N/A	N/A	N/A	N/A	N/A	+0.68	N/A	N/A	N/A	N/A

Nomenclature: See separate page